Day 13 — Genome Annotation

BIOL 4810

2026-03-08

Day 13 — Genome Annotation + Species Identification

Goal: Use **Prokka** (Galaxy) to annotate a bacterial genome assembly, interpret the output files, extract the **16S rRNA** for a quick BLAST-based ID, and then use **TYGS** + **iTOL** to place your genome on a phylogenomic tree.
Submission: Fill in the answers + upload required screenshots/files below → click **Prepare submission preview** → click **Download as PDF (Print)**.

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0) Files you need today

You should already have (from Day 12): - SPAdes assembly contigs file (FASTA)

0A. Assembly file name (as it appears in Galaxy History):

SPAdes.fasta

Local folder (for Git later)

Create a folder inside your BIOL 4810 course folder, for example: - `module2_wgs/day13_prokka_tygs/`

0B. Local folder path/description:

Desktop/biol4810/module02/day13_prokka_tygs

0C) Filter contigs for Prokka (≥500 bp)

Some downstream tools struggle with very short contigs or very large numbers of contigs.

In Galaxy: 1. Search for the tool **Filter FASTA** (it may appear as *Filter sequences by length*). 2. Input: your original SPAdes **contigs FASTA**. 3. Set **minimum sequence length = 500 bp**. 4. Run the tool to create a **filtered contigs** dataset.

Use the filtered contigs for Prokka today.

Keep the original (unfiltered) contigs for TYGS. TYGS can take a long time and you already submitted the full assembly there.

0C. Filtered contigs dataset name (Galaxy History):

Galaxy2-[SPAdes.fasta].fasta

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1) Genome QC (CheckM2 in Galaxy)

Today you will start by evaluating your assembly quality in **Galaxy** using **CheckM2**, which estimates **completeness** and **contamination**.

1A — CheckM2

In Galaxy: 1. Run **CheckM2** on the **filtered assembly FASTA** you will use for Prokka. 2. Open the main CheckM2 results table/report.

CheckM2 tool version (from Job Info):

Galaxy Version 1.1.0+galaxy0

Completeness (%):

100.0

Contamination (%):

1.53

Interpretation (2–3 sentences): Would you trust this assembly for annotation/species ID? Why?

Yes, I would trust this assembly for an annotation/species ID because it is 100% complete and is only 1.53% contaminated. I feel that it is relatively clean and would be hard to get a 0% contamination, so a 1.53% contamination looks pretty good.

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2) Run Prokka on Galaxy

In Galaxy: 1. Search tools for **Prokka** 2. Input: your SPAdes **contigs FASTA** 3. Use defaults unless instructed otherwise 4. Run Prokka

Record your Prokka settings

2A. Prokka version (from Job Info):

Galaxy Version 1.14.6+galaxy1

2B. Any settings changed from defaults? If yes, what?

Defaults

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3) Prokka outputs — what is each file for?

From Prokka outputs in Galaxy, you will use these most today:

- **.txt** summary statistics
- **.gff** master annotation (features + coordinates)
- **.ffn** nucleotide sequences of gene features (includes rRNA genes)
- **.faa** amino acid sequences of predicted proteins

3A. In your own words, what is the difference between `.gff`, `.ffn`, and `.faa`? (2–4 sentences)

.gff: Tells you where everything is
.ffn: Gives the DNA sequences
.faa: The protein sequences.

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4) Interpret the Prokka summary (.txt)

Open the Prokka **.txt** summary.

4A. Organism line (paste exactly as reported — it may be "Unknown"):

organism: Genus species strain

4B. contigs:

contigs: 1802

4C. bases:

bases: 5261250

4D. CDS:

CDS: 4634

3E. rRNA (total rRNA features):

rRNA: 6

3F. tRNA:

tRNA: 75

3G. tmRNA:

tmRNA: 1

3H. Based on genome size and CDS count, does this look like a typical bacterial genome? Why/why not? (2–3 sentences)

The genome size came out to 5261250 bases, while the CDS count is 4634, meaning yes - this does look like the typical bacterial genome. Typically, bacterial genomes have anywhere between 4-6 MB and somewhere between 3500-5500 CDS. Rounded our bases to everyone 1000000 bases = 1MB, we can see that our genome has about

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5) rRNA features (16S/23S/5S) — paste evidence from Prokka

The Prokka `.txt` summary often reports only a total count (e.g., **rRNA: 3**) without listing 16S vs 23S vs 5S in that summary.

To see which rRNAs were found: - Open the **.gff** file and search for `rRNA`, `16S`, `23S`, `5S` **OR** - Open the **.ffn** file and search for `ribosomal RNA`

5A. Paste the description line (header or attribute text) for the 16S rRNA feature:

NODE_56_length_1670_cov_388.281858 barrnap:0.9 rRNA 51 1588 0 - .
ID=DICEOMCD_04562;locus_tag=DICEOMCD_04562;product=16S ribosomal RNA

5B. Paste the description line (header or attribute text) for the 23S rRNA feature:

NODE_53_length_1749_cov_386.573565 barrnap:0.9 rRNA 4 1724 0 + .
ID=DICEOMCD_04555;locus_tag=DICEOMCD_04555;note=aligned only 53 percent of the 23S ribosomal

5C. Paste the description line (header or attribute text) for the 5S rRNA feature:

NODE_352_length_286_cov_50.990431 barrnap:0.9 rRNA 201 280 2.5e-08 + .
ID=DICEOMCD_04620;locus_tag=DICEOMCD_04620;note=aligned only 67 percent of the 5S ribosomal

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6) Hypothetical proteins — estimate and interpret

Hypothetical proteins typically appear as: - `product=hypothetical protein` in `.gff` - “hypothetical protein” in headers/annotations in `.faa`

6A. How did you estimate or count hypothetical proteins? (1–2 sentences)

I searched for "hypothetical protein" and then looked at the number of hits that came up, which was 1190.

6B. Estimated number of hypothetical proteins:

1190

6C. Estimated % hypothetical proteins (of CDS):

1190/4634 = 0.257 = 25.7%

6D. What does a high % hypothetical proteins mean (and NOT mean)? (2–3 sentences)

A high amount of hypothetical proteins mean that there is no strong homology, it could be a novel sequence, or there is a limitation in the database. Just because the proteins are hypothetical, that doesn't mean they're junk, we just don't know what they do.

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7) Extract 16S rRNA from the Prokka .ffn

Open the Prokka **.ffn** file. Search for a header/annotation indicating **16S ribosomal RNA**.

7A. Paste the FASTA header line for the 16S entry you used:

>DICEOMCD_04562 16S ribosomal RNA

7B. Paste the full 16S rRNA sequence you used for BLAST (sequence only; you can keep line breaks):

ATGCAACGCGAAGAACCTTACCTGGTCTTGACATCCACAGAAGAATCCAGAGATGGATTT
GTGCCTTCGGGAACTGTGAGACAGGTGCTGCATGGCTGTCGTCAGCTCGTGTTGTGAAAT
GTTGGGTTAAGTCCCGCAACGAGCGCAACCCTTATCCTTTGTTGCCAGCGATTAGGTCGG
GAACTCAAAGGAGACTGCCAGTGATAAACTGGAGGAAGGTGGGGATGACGTCAAGTCATC
ATGGCCCTTACGACCAGGGCTACACACGTGCTACAATGGCGCATACAAAGAGAAGCGACC
TCGCGAGAGCAAGCGGACCTCATAAAGTGCGTCGTAGTCCGGATTGGAGTCTGCAACTCG
ACTCCATGAAGTCGGAATCGCTAGTAATCGTGGATCAGAATGCCACGGTGAATACGTTCC
CGGGCCTTGTACACACCGCCCGTCACACCATGGGAGTGGGTTGCAAAAGAAGTAGGTAGC
TTAACCTTCGGGAGGGCGCTTACCACTTTGTGATTCATGACTGGGGTGAAGTCGTAACAA
GGTAACCGTAGGGGAACCTGCGGTTGGATCACCTCCTT

7C. In 1 sentence: why do we use 16S for a quick ID?

The 16S is a conserved region that is found in all bacteria.

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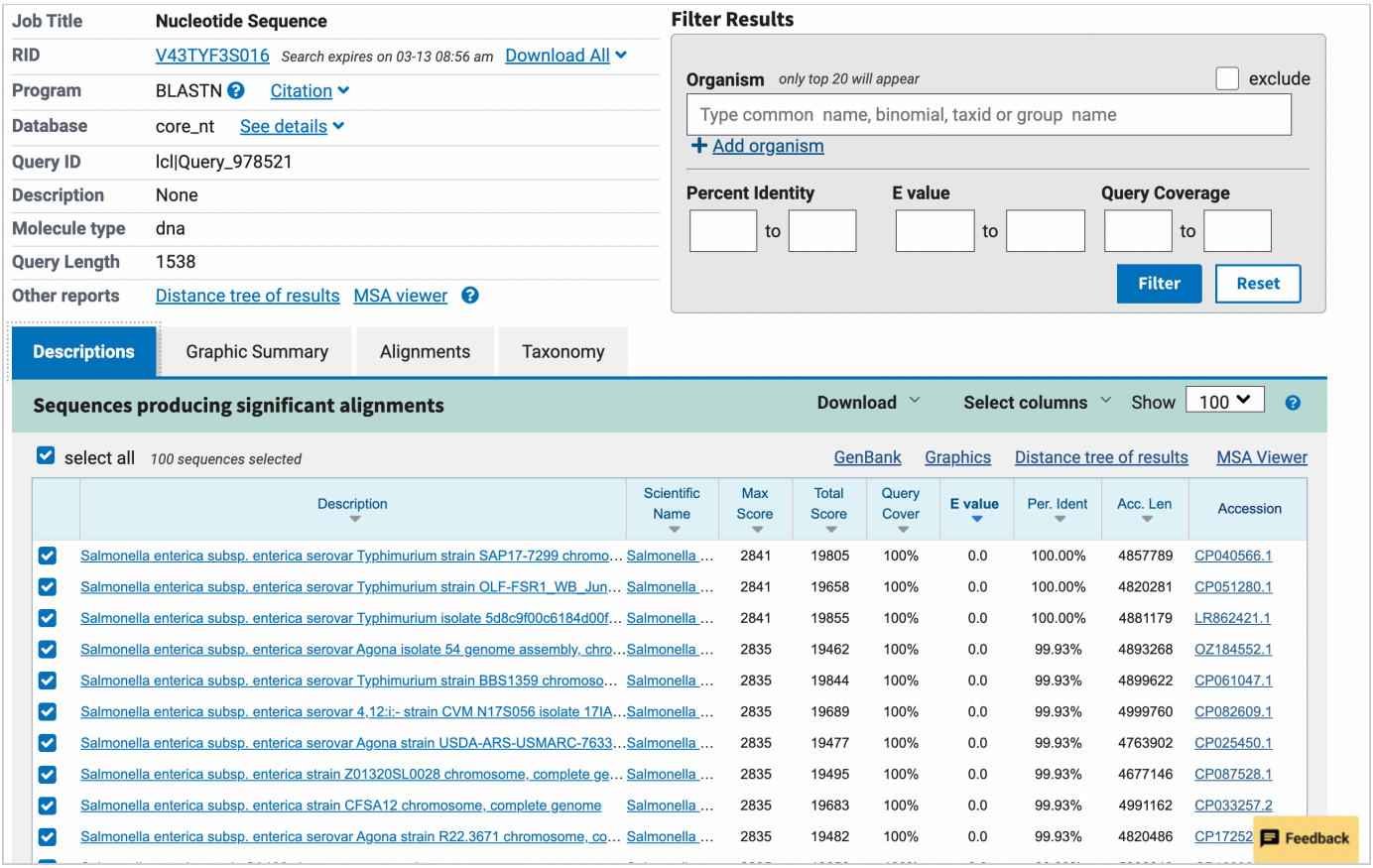
8) 16S BLAST (species ID quick check)

BLAST your 16S sequence: 1. Go to NCBI BLAST 2. Choose **BLASTn** 3. Paste your 16S sequence 4. Run BLAST and choose a reasonable hit to interpret

Upload screenshot: your 16S BLAST results (Descriptions table or Alignment view)

Choose File Screen Shot… 5.58.06 PM



Job Title	Nucleotide Sequence
RID	V43TYF3S016 Search expires on 03-13 08:56 am Download All
Program	BLASTN Citation
Database	core_nt See details
Query ID	lcl|Query_978521
Description	None
Molecule type	dna
Query Length	1538
Other reports	Distance tree of results MSA viewer

Filter Results

Organism only top 20 will appear exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to Query Coverage to E value to

Filter Reset

Descriptions Graphic Summary Alignments Taxonomy

Sequences producing significant alignments Download Select columns Show 100

select all 100 sequences selected GenBank Graphics Distance tree of results MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Salmonella enterica subsp. enterica serovar Typhimurium strain SAP17-7299 chromo…	Salmonella …	2841	19805	100%	0.0	100.00%	4857789	CP040566.1
Salmonella enterica subsp. enterica serovar Typhimurium strain OLF-FSR1_WB_Jun…	Salmonella …	2841	19658	100%	0.0	100.00%	4820281	CP051280.1
Salmonella enterica subsp. enterica serovar Typhimurium isolate 5d8c9f00c6184d00f…	Salmonella …	2841	19855	100%	0.0	100.00%	4881179	LR862421.1
Salmonella enterica subsp. enterica serovar Agona isolate 54 genome assembly, chro…	Salmonella …	2835	19462	100%	0.0	99.93%	4893268	OZ184552.1
Salmonella enterica subsp. enterica serovar Typhimurium strain BBS1359 chromoso…	Salmonella …	2835	19844	100%	0.0	99.93%	4899622	CP061047.1
Salmonella enterica subsp. enterica serovar 4,12:i:- strain CVM N17S056 isolate 17IA…	Salmonella …	2835	19689	100%	0.0	99.93%	4999760	CP082609.1
Salmonella enterica subsp. enterica serovar Agona strain USDA-ARS-USMARC-7633…	Salmonella …	2835	19477	100%	0.0	99.93%	4763902	CP025450.1
Salmonella enterica subsp. enterica strain Z01320SL0028 chromosome, complete ge…	Salmonella …	2835	19495	100%	0.0	99.93%	4677146	CP087528.1
Salmonella enterica subsp. enterica strain CFSA12 chromosome, complete genome	Salmonella …	2835	19683	100%	0.0	99.93%	4991162	CP033257.2
Salmonella enterica subsp. enterica serovar Agona strain R22.3671 chromosome, co…	Salmonella …	2835	19482	100%	0.0	99.93%	4820486	CP17252

Feedback

8A. Top hit organism name (as shown):

Salmonella enterica sybsp. enterica serovar Typhimurium

8B. % identity and query coverage for that hit:

100.00% and 100%

8C. Your best interpretation of species/genus (be cautious) (2–3 sentences):

Based on the percent identity being 100.00% and query coverage being 100%, we can assume that the our mystery organism belongs to the Salmonella enterica sybsp. enterica serovar Typhimurium group. Once again, this is only an assumption as there is always the possibility that something was done wrong. The 16s is also a

8D. One reason 16S BLAST can be misleading or insufficient (1–2 sentences):

As previously mentioned, a 16S BLAST is looking at a single genome and making a single comparison on that.

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9) ANI confirmation (EZBioCloud ANI)

After you have a likely species ID from **16S BLAST**, use ANI as a stronger genome-wide comparison.

You will use the **EZBioCloud ANI** tool here (open in a new tab): https://www.ezbiocloud.net/tools/ani (https://www.ezbiocloud.net/tools/ani)

What you will do 1. Download your **FILTERED contigs FASTA** (the same one you used for Prokka today). 2. Download **two reference genomes** (FASTA) from NCBI for *two candidate species* you want to compare against (often the top 16S BLAST hit and one close alternative). 3. Run ANI twice (your genome vs reference genome 1; your genome vs reference genome 2).

9A. Species/strain for comparison #1 (your chosen reference):

S. Typhimurium

9B. ANI for comparison #1 (%):

99.94

9C. Species/strain for comparison #2 (your chosen reference):

S. Bongori

9D. ANI for comparison #2 (%):

89.82

9E. Interpretation (2–3 sentences): Which comparison supports the strongest species-level match, and why?

Between S. Bongori and S. Typhimurium, S. Typhimurium is the closest match. This is because S. Typhimurium came out with an ANI value of 99.94, while S. Bongori had an ANI value of 89.82.

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10) TYGS — what it provides

Submit your assembly to TYGS (Type Strain Genome Server). TYGS will: - Compare your genome to type strains - Provide **dDDH** (species boundary ~70%) - Produce a phylogenomic tree

10A. In your own words: what is TYGS doing that 16S BLAST is not? (2–4 sentences)

TYGS compares your genome to type strains while 16S is only a single genome, making TYGS much stronger than 16S alone.

10B. Record your closest type strain match (name) and dDDH value (if available today):

Salmonella enterica LT2 & 99.5

10C. What does the 70% dDDH threshold mean? (1–2 sentences)

It determines how related they are. So, a 70% or higher dDDH threshold allows us to confidently believe that the two species are related.

10D. If TYGS shows low branch support and offers a proteome-based tree, what does that imply? (2–3 sentences)

If TYGS was to show low branch support, then that means it's confidence in that branch is low. In comparison, a branch with high support signifies a branch with high confidence. If a proteome tree is offered, it is because the TYGS does not feel very confident in the genome tree it created, so the Proteome tree can offer more data to

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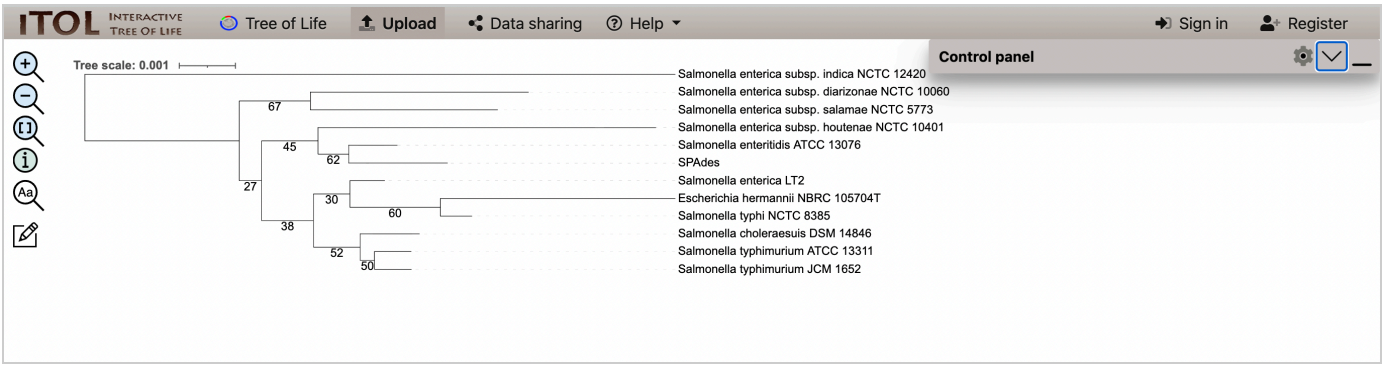
11) iTOL — visualize your TYGS tree

1. From TYGS, download the **tree file** (**.phy**)
2. Go to **iTOL** (Interactive Tree of Life)
3. Upload the tree
4. Adjust labels/formatting enough to be readable
5. Take a screenshot

Upload screenshot: iTOL tree visualization (from TYGS tree)

Choose File Screen Shot… 9.47.10 AM



11A. What does the tree tell you that a single BLAST hit does not? (2–3 sentences)

The tree helps explain the assumed relationship between the different Salmonella species. This relationship is only assumed based on previous patterns.

11B. Identify one branch you trust and one you would be cautious about. Explain using support values if shown. (2–4 sentences)

I would trust the branch labeled with a 67 and would not trust the branch labeled with 27. The larger the bootstrap, the more statistically confident the tree is in its interpretation.

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12) What to download and save for Git

Minimum recommended downloads for your local folder: - Prokka **.txt** summary - Prokka **.gff** - Prokka **.ffn** - Prokka **.faa** - CheckM2 results/report (table) - TYGS tree file (**.phy**) - Your PDF from this page (print-to-PDF)

12A. Write 4–6 lines of a "workflow note" you would put in `workflow_day13.md` (include tool versions when possible).

These were the notes I took in order to upload stuff to GitHub

How to Upload to GitHub Using Terminal
Step 1) "cd *drag biol4810 folder into terminal*"
Step 2) "git status"
Step 3) "git add *click and drag selected folder*"

Export (PDF)

1. Upload the required screenshots above (BLAST + iTOL)
2. Click **Prepare submission preview**
3. Click **Download as PDF (Print)**

Submission preview

Setup
Assembly file: SPAdes.fasta
Local folder: Desktop/biol4810/module02/day13_prokka_tygs

Prokka run
Version: Galaxy Version 1.14.6+galaxy1
Settings: Defaults

Outputs understanding
.gff vs .ffn vs .faa: .gff: Tells you where everything is .ffn: Gives the DNA sequences .faa: The protein sequences.

Prokka summary (.txt)
Organism: organism: Genus species strain
Contigs: contigs: 1802 *Bases:* bases: 5261250
CDS: CDS: 4634 *rRNA:* rRNA: 6 *tRNA:* tRNA: 75 *tmRNA:* tmRNA: 1
Sanity check: The genome size came out to 5261250 bases, while the CDS count is 4634, meaning yes - this does look like the typical bacterial genome. Typically, bacterial genomes have anywhere between 4-6 MB and somewhere between 3500-5500 CDS. Rounded our bases to everyone 1000000 bases = 1MB, we can see that our genome has about 5MB and it falls in between that 3500-5500 CDS range at 4634 CDS.

rRNA evidence
16S description: NODE_56_length_1670_cov_388.281858 barrnap:0.9 rRNA 51 1588 0 - . ID=DICEOMCD_04562;locus_tag=DICEOMCD_04562;product=16S ribosomal RNA
23S description: NODE_53_length_1749_cov_386.573565 barrnap:0.9 rRNA 4 1724 0 + . ID=DICEOMCD_04555;locus_tag=DICEOMCD_04555;note=aligned only 53 percent of the 23S ribosomal RNA;product=23S ribosomal RNA (partial)
5S description: NODE_352_length_286_cov_50.990431 barrnap:0.9 rRNA 201 280 2.5e-08 + . ID=DICEOMCD_04620;locus_tag=DICEOMCD_04620;note=aligned only 67 percent of the 5S ribosomal RNA;product=5S ribosomal RNA (partial)

CheckM2
CheckM2 version: Galaxy Version 1.1.0+galaxy0
Completeness: 100.0% *Contamination:* 1.53%
CheckM2 interpretation: Yes, I would trust this assembly for an annotation/species ID because it is 100% complete and is only 1.53% contaminated. I feel that it is relatively clean and would be hard to get a 0% contamination, so a 1.53% contamination looks pretty good.

Hypothetical proteins
How estimated: I searched for "hypothetical protein" and then looked at the number of hits that came up, which was 1190.
 # hypothetical: 1190 *% hypothetical:* 1190/4634 = 0.257 = 25.7%
Interpretation: A high amount of hypothetical proteins mean that there is no strong homology, it could be a

novel sequence, or there is a limitation in the database. Just because the proteins are hypothetical, that doesn't mean they're junk, we just don't know what they do.

16S extraction

16S header: >DICEOMCD_04562 16S ribosomal RNA

16S sequence:

```
TTGAAGAGTTTGATCATGGCTCAGATTGAACGCTGGCGGCAGGCCTAACACATGCAAGTC
GAACGGTAACGGGAAGCAGCTTGCTGCTTCGCTGACGAGTGGCGGACGGGTGAGTAATGT
CTGGGAAACTGCCTGATGGAGGGGGATAACTACTGGAAACGGTGGCTAATACCGCATAAC
GTCGCAAGACCAAAGAGGGGGACCTTCGGGCCTCTTGCCATCAGATGTGCCCAGATGGGA
TTAGCTTGTTGGTGAGGTAACGGCTCACCAAGGCGACGATCCCTAGCTGGTCTGAGAGGA
TGACCAGCCACACTGGAACTGAGACACGGTCCAGACTCCTACGGGAGGCAGCAGTGGGGA
ATATTGCACAATGGGCGCAAGCCTGATGCAGCCATGCCGCGTGTATGAAGAAGGCCTTCG
GGTTGTAAAGTACTTTCAGCGGGGAGGAAGGTGTTGTGGTTAATAACCGCAGCAATTGAC
GTTACCCGCAGAAGAAGCACCGGCTAACTCCGTGCCAGCAGCCGCGGTAATACGGAGGGT
GCAAGCGTTAATCGGAATTACTGGGCGTAAAGCGCACGCAGGCGGTCTGTCAAGTCGGAT
GTGAAATCCCCGGGCTCAACCTGGGAACTGCATTCGAAACTGGCAGGCTTGAGTCTTGTA
GAGGGGGGTAGAATTCCAGGTGTAGCGGTGAAATGCGTAGAGATCTGGAGGAATACCGGT
GGCGAAGGCGGCCCCCTGGACAAAGACTGACGCTCAGGTGCGAAAGCGTGGGGAGCAAAC
AGGATTAGATACCCTGGTAGTCCACGCCGTAAACGATGTCTACTTGGAGGTTGTGCCCTT
GAGGCGTGGCTTCCGGAGCTAACGCGTTAAGTAGACCGCCTGGGGAGTACGGCCGCAAGG
TTAAAACTCAAATGAATTGACGGGGGCCCGCACAAGCGGTGGAGCATGTGGTTTAATTCG
ATGCAACGCGAAGAACCTTACCTGGTCTTGACATCCACAGAAGAATCCAGAGATGGATTT
GTGCCTTCGGGAACTGTGAGACAGGTGCTGCATGGCTGTCGTCAGCTCGTGTTGTGAAAT
GTTGGGTTAAGTCCCGCAACGAGCGCAACCCTTATCCTTTGTTGCCAGCGATTAGGTCGG
GAACTCAAAGGAGACTGCCAGTGATAAACTGGAGGAAGGTGGGGATGACGTCAAGTCATC
ATGGCCCTTACGACCAGGGCTACACACGTGCTACAATGGCGCATACAAAGAGAAGCGACC
TCGCGAGAGCAAGCGGACCTCATAAAGTGCGTCGTAGTCCGGATTGGAGTCTGCAACTCG
ACTCCATGAAGTCGGAATCGCTAGTAATCGTGGATCAGAATGCCACGGTGAATACGTTCC
CGGGCCTTGTACACACCGCCCGTCACACCATGGGAGTGGGTTGCAAAAGAAGTAGGTAGC
TTAACCTTCGGGAGGGCGCTTACCACTTTGTGATTCATGACTGGGGTGAAGTCGTAACAA
GGTAACCGTAGGGGAACCTGCGGTTGGATCACCTCCTT
```

Why 16S: The 16S is a conserved region that is found in all bacteria.

16S BLAST

Top hit: Salmonella enterica sybsp. enterica serovar Typhimurium

Metrics: 100.00% and 100%

Interpretation: Based on the percent identity being 100.00% and query coverage being 100%, we can assume that the our mystery organism belongs to the Salmonella enterica sybsp. enterica serovar Typhimurium group. Once again, this is only an assumption as there is always the possibility that something was done wrong. The 16s is also a single genome, so a large assumption is being made using the single genome.

Limitation: As previously mentioned, a 16S BLAST is looking at a single genome and making a single comparison on that.

BLAST screenshot

Job Title	Nucleotide Sequence
RID	V43TYF3S016 *Search expires on 03-13 08:56 am* Download All
Program	BLASTN Citation
Database	core_nt See details
Query ID	lcl|Query_978521
Description	None
Molecule type	dna
Query Length	1538
Other reports	Distance tree of results MSA viewer

Filter Results

Organism *only top 20 will appear* exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to | E value to | Query Coverage to

Filter Reset

Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments Download Select columns Show 100

select all *100 sequences selected* GenBank Graphics Distance tree of results MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Salmonella enterica subsp. enterica serovar Typhimurium strain SAP17-7299 chromo...	Salmonella ...	2841	19805	100%	0.0	100.00%	4857789	CP040566.1
Salmonella enterica subsp. enterica serovar Typhimurium strain OLF-FSR1_WB_Jun...	Salmonella ...	2841	19658	100%	0.0	100.00%	4820281	CP051280.1
Salmonella enterica subsp. enterica serovar Typhimurium isolate 5d8c9f00c6184d00f...	Salmonella ...	2841	19855	100%	0.0	100.00%	4881179	LR862421.1
Salmonella enterica subsp. enterica serovar Agona isolate 54 genome assembly, chro...	Salmonella ...	2835	19462	100%	0.0	99.93%	4893268	OZ184552.1
Salmonella enterica subsp. enterica serovar Typhimurium strain BBS1359 chromoso...	Salmonella ...	2835	19844	100%	0.0	99.93%	4899622	CP061047.1
Salmonella enterica subsp. enterica serovar 4,12:i:- strain CVM N17S056 isolate 17IA...	Salmonella ...	2835	19689	100%	0.0	99.93%	4999760	CP082609.1
Salmonella enterica subsp. enterica serovar Agona strain USDA-ARS-USMARC-7633...	Salmonella ...	2835	19477	100%	0.0	99.93%	4763902	CP025450.1
Salmonella enterica subsp. enterica strain Z01320SL0028 chromosome, complete ge...	Salmonella ...	2835	19495	100%	0.0	99.93%	4677146	CP087528.1
Salmonella enterica subsp. enterica strain CFSA12 chromosome, complete genome	Salmonella ...	2835	19683	100%	0.0	99.93%	4991162	CP033257.2
Salmonella enterica subsp. enterica serovar Agona strain R22.3671 chromosome, co...	Salmonella ...	2835	19482	100%	0.0	99.93%	4820486	CP17252

Feedback

TYGS

TYGS vs 16S: TYGS compares your genome to type strains while 16S is only a single genome, making TYGS much stronger than 16S alone.

Closest match + dDDH: Salmonella enterica LT2 & 99.5

70% threshold: It determines how related they are. So, a 70% or higher dDDH threshold allows us to confidently believe that the two species are related.

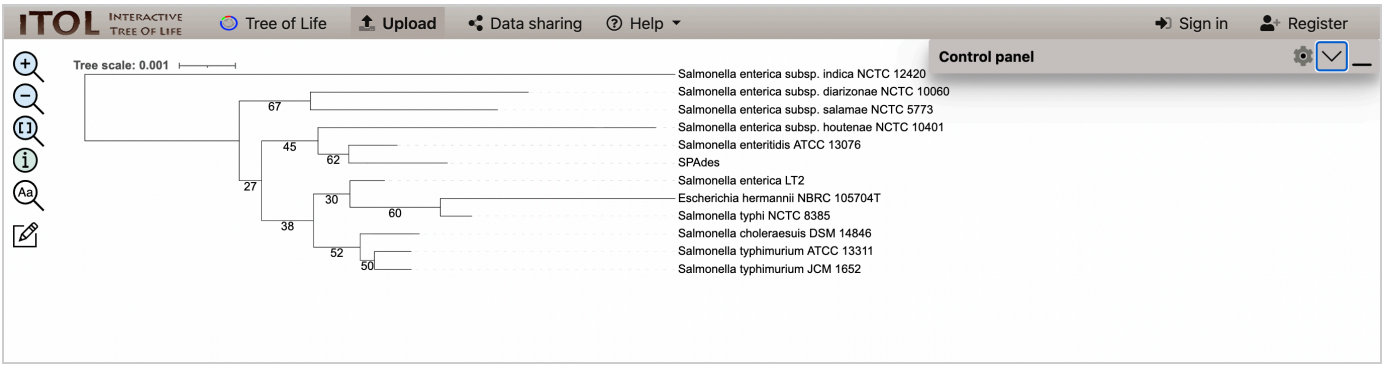
Proteome tree option (low support): If TYGS was to show low branch support, then that means it's confidence in that branch is low. In comparison, a branch with high support signifies a branch with high confidence. If a proteome tree is offered, it is because the TYGS does not feel very confident in the genome tree it created, so the Proteome tree can offer more data to reach a better conclusion.

iTOL tree

Tree adds value because: The tree helps explain the assumed relationship between the different Salmonella species. This relationship is only assumed based on previous patterns.

Support/trust: I would trust the branch labeled with a 67 and would not trust the branch labeled with 27. The larger the bootstrap, the more statistically confident the tree is in its interpretation.

iTOL tree screenshot



Workflow note (for Git)

```
These were the notes I took in order to upload stuff to GitHub

How to Upload to GitHub Using Terminal
Step 1) “cd *drag biol4810 folder into terminal*”
Step 2) “git status”
Step 3) “git add *click and drag selected folder*”
Step 4) “git status” -> this double checks its being done right.
Step 5) “git commit -m “uploading annotation folder from bioinformatics””
Step 6) “git push”
```